

Josiah Chen

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EDUCATION

University of California, Los Angeles (UCLA)

Expected June 2028

B.S. in Computer Science and Math; GPA: 3.98

Los Angeles, CA

Relevant Coursework: Data Structures and Algorithms, Computing in C++, Discrete Structures, Linear Algebra, Multivariable Calculus, Differential Equations, Probability

Awards: \$10,000 National Merit Scholarship, \$8,000 John Muir Health Scholarship, AIME Qualifier, USA Computing Olympiad (USACO) Silver

EXPERIENCE

Trustworthy AI Lab

Summer 2026

Software Engineering Intern

Los Angeles, CA

- Engineered a Python pipeline to automatically embed and verify provenance metadata on AI-generated audio and music from the ElevenLabs and Suno platforms, reducing manual verification effort.
- Automated compliance checks against California's AI Transparency Act by translating legal requirements into scripted validation logic, cutting manual review time for the compliance team.
- Built a PyTorch/NumPy test suite to benchmark watermark robustness against compression and other distortions, catching regressions before release.
- Designed, built, and shipped a public-facing web application for looking up and verifying the provenance of AI-generated content.

Association for Computing Machinery (ACM) - AI Division

Sep. 2025 - Present

Software Engineering Intern

Los Angeles, CA

- Designed, implemented, and debugged deep learning models (CNNs) in Python with PyTorch/TensorFlow, iterating on architecture and code structure to improve performance and maintainability.
- Implemented core algorithms (linear regression, gradient descent) from scratch in Python, writing unit tests to validate correctness against reference implementations.
- Translated research papers on generative model architectures, including diffusion forward/reverse processes, into working, well-documented production code.
- Collaborated with 5+ engineers in weekly code reviews and technical discussions, strengthening team practices around testing, version control, and reproducible builds.

PROJECTS

Audiomark AI | Python, Next.js, React, Vite, Claude Code, Codex

August 2026

- Architected and shipped a full-stack compliance MVP, extending validation work from a Trustworthy AI Lab internship project, for verifying AI-generated audio watermarks, system provenance, and no-retention data handling.
- Built a Python validation harness to test watermark detection accuracy and robustness against audio manipulation.
- Developed a public-facing lookup tool with a React/Vite and Next.js frontend for verifying AI-generated audio and tracing provenance metadata end to end.
- Used Claude Code and Codex to accelerate development and testing, and deployed the app to production on Vercel with a no-retention architecture.

AI Image Generator | Python, TensorFlow, NumPy, ImageIO, Matplotlib

June 2024

- Engineered a Deep Convolutional GAN (DCGAN) in Python as a modular, reusable codebase to generate realistic digit images.
- Built an end-to-end training pipeline in TensorFlow with NumPy for numerical operations and PIL/ImageIO for image I/O and preprocessing.
- Tuned generator and discriminator network architectures, engineering training stability for balanced adversarial learning.
- Instrumented training with Matplotlib visualizations to monitor convergence and accelerate debugging.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, HTML/CSS

Software Engineering: Data Structures & Algorithms, Object-Oriented Design, Testing, Code Review, Version Control (Git)

Frameworks & Frontend: React, Next.js, Vite, PyTorch, TensorFlow

Tools & Platforms: Git, GitHub, GitHub Pages, Vercel, VS Code, Visual Studio, Google Colab, Claude Code, Codex

Libraries: NumPy, pandas, Matplotlib, ImageIO

Spoken Languages: English, Mandarin, Spanish